Hannah T. Nguyen

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EDUCATION

The University of Tulsa — Tulsa, Oklahoma
M.S. Electrical Engineering — May 2028 (expected)
B.S. Electrical and Computer Engineering — May 2027, GPA: 4.000
Minors in Mathematics and Computer Science

Union High School — Tulsa, Oklahoma
High School Diploma — May 2023, GPA: 4.96 (Weighted)

WORK EXPERIENCE

U.S. Government — **Washington D.C.**
Digital Engineering Intern — Summer 2026

- Created an enterprise-wide database tracking interface for digital forensics, streamlining legacy workflows for approvals, procurement, and renewals, improving productivity, and was presented to senior leadership.
- Designed and developed the web application using HTML, CSS, Javascript, Bootstrap, Python, and Flask, later migrating it to Node.js and Angular to improve scalability, maintainability, and performance, while using Ansible for deployment and configuration management.
- Performed hardware teardowns and comparative analysis of similar consumer electronic and electromechanical devices from multiple manufacturers, including data extraction and disassembly/reassembly of mobile, laptop, and other select hardware.
- Documented detailed technical documentation of teardowns, device comparisons, and project findings, including extensive documentation for an individual intern project.
- Communicated regularly with management to report progress, address challenges and roadblocks, and incorporate feedback for continuous improvement.

University School, University of Tulsa — **Tulsa, Oklahoma**
Lab Assistant — Fall 2025 – Present

- Presented hands-on projects to middle school computer science and engineering classes, promoting STEM engagement through interactive learning experiences.
- Assisted in managing a multi-node Raspberry Pi rack, supporting system setup, maintenance, and experimentation with distributed computing environments.

U.S. Government — **Washington D.C.**
Software Developer Intern — Summer 2025

- Developed a full-stack application from scratch to support AI-enabled triage, retrieval, and exploitation of sensitive data from complex technical collections, rapidly learning and implementing a new framework.
- Used Vue3 and Vuetify to build an intuitive user interface and AI chatbot, integrating AI API calls for efficient search and retrieval, while improving usability through structured data presentation and streamlined functionality.
- Collaborated with cross-functional teams including data scientists, engineers, and IT administrators to support AI/ML integration, cloud application development, and mission-focused data workflows.
- Improved system reliability and maintainability by debugging, cleaning, and documenting code, while communicating technical progress and issues across teams.
- Recognized for strong performance, technical capability, and ability to integrate effectively within a high-performing team.

NCFI Laboratory, University of Tulsa — **Tulsa, Oklahoma**
Undergraduate Researcher — Fall 2023 – May 2025

- Conducted research on the development and applications of alternative reality software and tools in order to quickly and affordably re-create digital versions of crime scenes as well as reconstruction large areas to allow for digital analysis of protection locations during event planning.
- Co-authored a white paper titled *Crime Scene Recreation in VR*, evaluating low-cost methods for reconstructing physical crime scenes at 1:1 scale and exploring remote VR collaboration for law enforcement investigations and training.
- Co-authored a white paper titled *Rokid MAX AR Glasses Penetration Analysis*, documenting security analysis of Android-based AR glasses through firmware modification attempts, hardware investigation, and chip-off techniques for extracting and analyzing flash memory.

TURC Program, University of Tulsa **Tulsa, Oklahoma**
Volunteer Java Programming Instructor Summer 2024

- Co-taught two Java programming classes to 37 incoming freshmen.
- The courses gave the students a head start in their computer science education, enabling them to bypass the first two classes, Introduction to Programming and Problem Solving and Fundamentals of Algorithms and Computer Applications, and take advanced courses during their first semester in college.

Junior TURC Program, University of Tulsa **Tulsa, Oklahoma**
High School Researcher Summer 2022

- Helped update the Rayzor Hall keycard system from an Arduino to an STMicroelectronics implementation.
- Redesigned 3 PCBs with KiCad for physical adaptation of the game "Keep Talking and Nobody Explodes."

Information Technology Department, Union Public Schools **Tulsa, Oklahoma**
Information Technology Intern Fall 2022 – Spring 2023

- Provided in-person assistance to students with information technology problems.
- Repaired laptop hardware and installed software.

SKILLS & PROFICIENCY

- **Electrical & Embedded:** Hardware reverse engineering, circuit analysis, electrical wiring, Raspberry Pi, Arduino, STM32, Altium Designer, KiCad, VHDL, STM32, soldering, HackRF, URH
- **Programming:** Python, Java, C, C++, JavaScript, HTML, CSS, Vue, Vuetify3, MIPS, Git, Lua, NodeJS, Angular
- **Engineering Software:** MATLAB, Simulink, Simscape, SOLIDWORKS, Fusion 360, Autodesk Inventor, VS Code, Ubuntu, LaTeX, SPICE, Xilinx, SDR++, GNURadio
- **Professional:** Project management, technical writing, product testing, independent research, leadership, Vietnamese, Microsoft Office 365, Google Professional Suite

PROJECTS

FPGA Deep-Capture Logic Analyzer Senior Design Project With Analog Devices Aug. 2026 - May 2027

- Architecting a deep-capture FPGA logic analyzer/data logger with configurable synchronous/asynchronous sampling, signal grouping, change-based compression, and DRAM-backed waveform storage.
- Designing a companion FPGA front-end for acquisition of 1.2-3.3 V external digital signals and high-speed serialized transfer to the main FPGA over LVDS-class interfaces.
- Developing waveform export and host connectivity for analysis through standard waveform tools such as VCD-compatible viewers, PulseView, or ngscopeclient.

Radio Frequency Signal Reverse Engineering Fall 2023

- Conducted a capstone research project on reverse engineering University of Tulsa ID card systems, using Proxmark3 to read and analyze cards and identifying three software and hardware vulnerabilities.
- Co-authored a white paper titled *Radio Frequency Signal Reverse Engineering: University of Tulsa ID Analysis*, documenting identified vulnerabilities and proposing security mitigations including password protection with lock bits, reinforced hardware design, and card number obfuscation.
- Course covers software-defined radios, radio frequency signal generation, resiliency techniques, signal analysis, reverse engineering and exploitation, radar basics, electronic warfare and cyber operations.
- Projects involve signal reversing and modulation of a radio-controlled toy, creating GNU Radio blocks, signal reversing of obfuscated communications, GPS signal exploitation, ADS-B signal exploitation and NFC signal exploitation.

Hardware Reverse Engineering Spring 2025

- Capstone project involved reverse engineering a custom, unknown PCB and implemented the functionality on an FGPA board using VHDL
- Course covers basic circuit theory, soldering, microcontrollers, field programmable gate arrays (FPGAs), serial communications, computer architectures and bootloaders.
- Projects involve circuit design, microcontroller programming, FPGAs and board-level reverse engineering.

STM32N6 Embedded AI UART Interface Spring 2026

- Designed a PC-to-STM32N6 embedded AI pipeline in C and Python to stream webcam frames over UART to a NUCLEO-N657X0-Q for object-detection processing and return detection results to a desktop GUI.
- Implemented a custom binary communication protocol with frame metadata and CRC32 validation, including ring-buffer packet parsing, RGB565-to-RGB888 conversion, and packetized detection responses on the STM32.
- Integrated RTOS-based task scheduling to coordinate UART communication, image processing, and AI inference, enabling efficient task execution and synchronization.

- Developed a Python Tkinter and OpenCV GUI to capture and resize webcam frames, communicate with the microcontroller over serial, parse returned detections, and overlay bounding boxes and inference results on live video.

IPS Display Module Hardware Design Fall 2025

- Designed schematic and PCB for a custom 2" IPS display module in Altium Designer, integrating an STM32G0B1CEU MCU and ER-TFT0.99-2 display in accordance with Hexabitz design requirements.

Personal Portfolio Website

- Project involved designing and deploying a personal portfolio website using Vue3 framework hosted on Github.
- Implemented responsive layouts, interactive features, and version control with Git to showcase projects and skills.

Projectile Trajectory Simulation Fall 2025

- Project involved programming in MARS MIPS assembly and C to calculate projectile time-of-flight, maximum height, and trajectory angle based on user input.
- Applied physics equations of motion to model artillery firing scenarios, validated results across both implementations, and documented methodology, findings, and error analysis in a professional report.

FTC Robotics Autonomous Control System 2022 – 2023

- Served as the main software developer for an FTC competition robot, independently researching and implementing Java and Kotlin control software on the REV Robotics Control Hub for autonomous and driver-controlled operation.
- Researched, implemented, and tuned PID-based trajectory following using RoadRunner, mecanum drive, wheel encoders, and dead-wheel odometry to improve autonomous localization and path accuracy.
- Researched and integrated OpenCV-based computer vision, sensors, drivetrain localization, and autonomous mechanisms while debugging hardware and software issues affecting robot motion and positioning.

3D Print Defect Detection Using STM32 and AI Spring 2026

- Developed an AI-based defect detection system to classify 3D prints as good or defective using an STM32 Nucleo, Python, and NanoEdge AI Studio.
- Built an automated image cleaning pipeline to provide consistent image inputs for the classification model and integrated communication between the Python application and STM32 hardware.
- Debugged embedded system constraints including model memory usage and packet communication failures, simplifying the AI model and implementing data integrity checks to improve reliable communication.

Single-Port Read/Write Memory Design Spring 2025

- Lab focused on designing and constructing a 16×4-bit memory device using a 64-bit shift register, address counter, and supporting logic.
- Implemented address comparison with XNOR/AND gates, clock division using T flip-flops, and read/write functionality through multiplexers and D flip-flops.
- Verified functionality with LEDs and 7-segment displays, and debugged timing and hardware issues to ensure reliable operation.

Custom Wattmeter Fall 2025

- Designed and constructed a low-cost digital wattmeter using an Arduino Uno R3, voltage-divider circuitry, a current-sense resistor, and an operational amplifier to measure voltage, current, and electrical power.
- Calibrated ADC measurements using linear regression to compensate for resistor and conversion errors, achieving less than 5% measurement error across tested operating points and documenting the design, calculations, testing, and results in a technical report.

SELECTED TECHNICAL PAPERS

Crime Scene Recreation in VR Summer 2024

- Co-authored a white paper evaluating low-cost methods for reconstructing physical crime scenes at 1:1 scale and exploring remote VR collaboration for law enforcement investigations and training.

Rokid MAX AR Glasses Penetration Analysis Spring 2024

- Co-authored a white paper documenting security analysis of Android-based AR glasses through firmware modification attempts, hardware investigation, and chip-off techniques for extracting and analyzing flash memory.

Radio Frequency Signal Reverse Engineering: University of Tulsa ID Analysis Fall 2023

- Co-authored a white paper documenting identified vulnerabilities and proposing security mitigations including password protection with lock bits, reinforced hardware design, and card number obfuscation.

CERTIFICATIONS

Certifications

SANS GIAC Incident Handler Certification (GCIH) January 2025

- GIAC Incident Handler (GCIH) certified professional with proven ability to detect, respond to, and resolve security incidents through deep understanding of attack techniques, vectors, tools, and effective incident response strategies.

SANS Global Industrial Cyber Security Professional Certification (GICSP) November 2025

- Global Industrial Cyber Security Professional (GICSP) certified, bridging IT, engineering, and cybersecurity to secure industrial control systems across their full lifecycle in vendor-neutral, operational environments.

TS/SCI Clearance

AWARDS & ACTIVITIES

IEEE TU Chapter – President and Honor Society Member (HKN) May 2024 – Present

- Led professional development workshops while engaging industry professionals for networking events.
- Increased ECE student participation in industry internship opportunities by 300%.
- Worked with the Department Chair to help improve the college.

Robotics Association of Tulsa Students (RATS) August 2026 – Present

- Embedded Systems and Mechanical Lead – National Underwater Robotics Team
- Responsible for creating and protecting electronics and systems underwater.
- Oversaw the main calculations for the construction of the robot.

Student Government Association – College of Engineering and Computer Science Senator September 2024 – January 2025

University of Tulsa President's Honor Roll January 2024 – Present

- Given to full-time students who maintain a term GPA of 4.0.

Alan Paller Cyber Scholarship January 2024 – Present

- Prestigious scholarship for excellence in cyber studies and research, and a commitment to serve with the federal government or OPM-approved organizations. The scholarship provides a stipend and the opportunity to take SANS courses and certification exams at no cost.

University of Tulsa Presidential Scholarship August 2023 – Present

- Full tuition scholarship awarded to high-achieving students who have demonstrated academic excellence in high school.

Oklahoma Academic Scholar May 2023

- Honor accorded to students who accumulate a minimum grade point average of 3.7 or were in the top 10% of their graduating class and obtained a minimum of 27 ACT composite score or a 1220 SAT score.

Tulsa Engineering Scholarship May 2023 – Present

- Scholarship awarded to high-achieving students who major in an engineering field.

FTC State Championship Control Award Sponsored by ARM, Inc. February 2023

- Awarded to high school teams that use sensors and software to increase robot functionality. Teams must demonstrate innovative thinking to solve challenges such as autonomous operation and improving mechanical systems using sensors and intelligent control schemes.